

W1 ML FOR BUSINESS

ML: learn mapping/structure from data to improve decisions at scale; next prediction, but action under constraints. **Supervised:** (x_i, y_i) for regression/classification; **unsup:** clusters, dimension reduction, density; **RL:** actions/rewards.

Problem framing: business decision → target Y , prediction time, unit of analysis, loss/cost, action threshold, feedback loop.

Pipeline: define objective → collect/data → split → EDA → features → train/tune → test once → deploy/monitor. **Leakage:** any feature unavailable at prediction time or computed using future/outcome.

Type	Goal	Metric
Reg	numeric $\hat{y}$	RMSE/MAE/ R^2
Class	$\hat{p}_c$	logloss/AUC/F1/cost
Rank	ordering	NDG/MAP
Cluster	segments	stability/silhouette

Exam cue: always state data, model, validation, metric, assumption, business cost.

W2 REGRESSION WORKFLOW

Supervised risk: $R(f) = E[L(Y, f(X))]$, empirical $\hat{R} = n^{-1} \sum_i L(y_i, f(x_i))$.

OLS: $\hat{\beta} = (X^T X)^{-1} X^T y$ if full rank; $\hat{y} = X \hat{\beta}$.

Linear regression: choose features, inspect residuals, fit by least squares/ML on Gaussian noise, validate on held-out data. **Assumptions** for inference: linear mean, iid errors, homoskedasticity, no severe collinearity.

KNN regression: $\hat{f}(x) = K^{-1} \sum_{i \in N_K(x)} y_i$; low K flexible/high variance, high K smooth/high bias. **Scale features** for distances.

Model	Strength	Failure
Linear	stable/interpretable	nonlinear signal
KNN	local/nonparam	high p , scale, slow pred
Transform	handles skew	harder interpretation

MSE decomposes into noise + bias² + variance; irreducible error cannot be trained away.

W3 DATA & FEATURES

Data-centric ML: improve labels, sampling, missingness, units, duplicates, time alignment before chasing complex models. **Feature engineering:** encode domain info into learnable signals.

Categoricals: one-hot $K-1$ for linear model w/ intercept; target encoding needs nested CV; rare levels: scale, winsorise only if justified, log/Box-Cox for skew, interactions for effect modification. Split before preprocessing fit: learn imputer/scaler/encoder on train only, then apply to val/test.

Missingness: MCAR/MAR/MNAR; add missing indicator if absence is informative. **Text/sparse:** counts/TF-IDF + NB/logistic/lasso.

Risk	Symptom	Fix
Leakage	too-good val	time split, pipeline
Shift	test drift	monitor PSI, recalibrate
Proxy bias	unfair action	audit protected proxies

Exam: say whether feature is available at decision time.

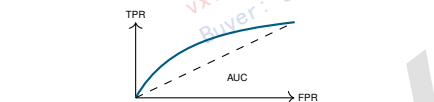
W4 CLASSIFICATION

Logistic: $p(y=1|x) = \sigma(\eta)$, $\eta = \beta_0 + x \beta$, odds multiply by e^{β_j} per unit. Logloss = $-\sum [y_i \log p_i + (1-y_i) \log(1-p_i)]$.

Bayes classifier: choose class minimizing expected cost with 0-1 loss choose max posterior. Gaussian DA: model $p(x|c)$ + prior $p(c)$; LDA shared covariance, QDA class covariance.

KNN classifier: local vote; $K=1$ memorizes, sensitive to noise/scale. **Metrics:** confusion matrix, TPR, FPR, precision, recall, specificity, F1, ROC-AUC, PR-AUC.

Threshold rule: predict positive iff $p > C_{FP}/(C_{FP} + C_{FN})$ when costs known.



Imbalance: cost can be useless; prefer PR-AUC/F1/cost-sensitive loss.

W5 COMPLEXITY & SELECTION

Generalisation: train error optimistic; validation estimates future risk; test used once.

Bias-variance: complexity ↑ usually bias ↓, variance ↑.

K-fold CV: split train into K folds, train on $K-1$, validate on held fold, average; nested CV for tuning + unbiased assessment.

Hyperparameter search: grid search, random better in high dimension, Bayesian uses surrogate/acquisition, early stopping saves budget. **Stacking:** out-of-fold base predictions train meta-learner; avoid fitting meta on in-sample predictions. **Model selection uncertainty:** chosen min CV error is noisy; use one-SE rule for simpler model.

Pattern	Diagnosis	Action
high train/high val underfit	low train/high val overfit	features/complexity
low train/high val overfit	unstable folds	regularise/data
unstable folds	variance/shift	stratify/group/time CV

W6 REGULARISATION

Penalised ERM: minimize $\sum_i L_i + \lambda C(\beta)$; λ trades fit vs complexity.

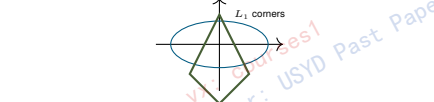
Ridge: $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$; shrinks correlated features together.

Lasso: $\sum_i L_i + \lambda |\beta|_1$; can set coefficients exactly 0; useful when feature search plausible.

Elastic net: $\lambda [(1-\alpha) |\beta|_2^2/2 + \alpha |\beta|_1]$; handles groups + sparsity. **Best subset:** combinatorial 2^p , unstable but interpretable.

Feature selection: filter (univariate), wrapper (CV search), embedded (lasso/step).

Post-selection OLS: may reduce bias but can increase variance and invalidate naive p-values.



Scale before regularising; otherwise penalty depends on units.

W7 DIAGNOSTIC TESTS

Diagnostics: baseline → learning curves → error slices → ablation → calibration → drift monitoring. **Learning curve:** if train/val both high, need regularisation/complexity; if gap large, need regularisation/data.

Calibration: predicted 0.8 should be right about 80%; check reliability plot/Brier; calibrate via Platt/isotonic. **Data split:** random for iid, time split for forecasting, group split for repeated customers/patients.

Deployment risk = data drift + concept drift + feedback loops + delayed labels + changing decision policy.

Failure	Cause	Test
Shortcut	proxy leakage	remove suspect vars
Slice error	minority segment	per-group metrics
Bad threshold	wrong costs	decision curve
Drift	new population	PSI/KL, monitor loss

Business answer: tie metric to action and cost, not abstract accuracy.

W8 NONLINEAR & GAMS

Basis expansion: $f(x) = \sum_j w_j b_j(x)$ makes nonlinear curves linear in parameters.

Spline = piecewise polynomial joined at knots; continuity constraints prevent jumps.

Smoothing spline: minimize $\sum_i (y_i - f(x_i))^2 + \lambda \int f''(t)^2 dt$; λ controls wiggleness. **Local regression:** weighted fit near x_0 ; bandwidth trades bias/variance.

GAM: $g(E[Y|X]) = \beta_0 + \sum_j f_j(X_j)$; interpretable additive nonlinear effects, limited interactions unless added.

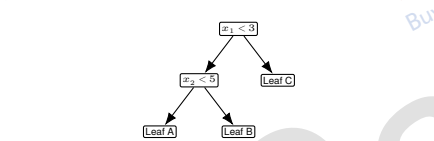
Method	Good for	Watch
Polynomial	global smooth	edge blow-up
Splines	local flex	knots/smoothness
GAM	explainable curves	interactions missed
KNN	local arbitrary	curse of dim

Curse: local methods need exponentially more data as p grows.

W9 TREES & FORESTS

CART: recursive binary splits choose variable/cut minimizing impurity. Regression impurity SSE; classification $Gini 1 - \sum p_k^2$ or entropy.

Tree prediction: follow splits to leaf, output mean/majority/probability. **Regularise tree:** max depth, min leaf, cost-complexity pruning. **CV Bagging:** bootstrap rows + average; variance ↓. **RF:** random split features decorrelate trees.



Model	Variance	Bias
Single tree	high	low-ish
Bagging	lower	similar
RF	lower if decorrelated	slightly higher

Importance bias: impurity importance favors high-cardinality variables; use permutation with caution under correlation.

W10 BOOSTING

Boosting: sequential learners focus on residuals/errors; additive model $F_M(x) = \sum_m \nu_m h_m(x)$. Gradient boosting: fit new tree to negative gradient of loss.

Squared loss: residual target $r_i = y_i - F_{m-1}(x_i)$; update $F_m = F_{m-1} + \nu h_m$.

AdaBoost: reweight misclassified observations, combine weak classifiers with weights: sensitive to label noise/outliers. **Tuning:** learning rate ν , number trees, depth, subsampling, min leaf.

Bagging	Boosting	Risk
parallel variance cut	sequential bias+variance cut	overfits if too many
deep trees ok	shallow often	noise sensitivity
		needs tuning

Small ν + many trees often strong; use validation/early stopping.

W11 MLE & TRAINING

Likelihood: $L(\theta) = p(D|\theta)$, log-likelihood $\ell = \sum_i \log p(y_i|x_i, \theta)$ for iid.

MLE $\hat{\theta} =$ arg max $\ell(\theta)$ = arg min $-\ell(\theta)$.

Gaussian regression: MLE for β equals least squares. **Bernoulli/logistic:** cross-entropy is negative log-likelihood; no closed form, optimise iteratively.

MAP: maximize log $p(D|\theta)$ + log $p(\theta)$; Gaussian prior gives ridge-like penalty. Laplace gives lasso-like.

Dist	NLL core	Use
Gaussian	$(y - \mu)^2$	regression binary
Bernoulli	$\log p - (1-y) \log(1-p)$	counts
Multinomial	$-\sum y_i \log p_i$	multi-class
Poisson	$-\mu - y \log \mu$	counts

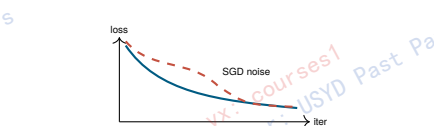
Always include independence/conditional model assumptions in likelihood answers.

W12 OPTIMISATION

Gradient descent: $\theta_{t+1} = \theta_t - \eta_t \nabla J(\theta_t)$; step size too large diverges, too small slow. **SGD:** noisy mini-batch gradients, cheaper and can help escape sharp regions.

Logistic gradient: $\nabla_\beta \ell = X^T (p - y)$ plus penalty term.

Second-order: Newton uses Hessian, fast near optimum but costly/unstable in high p . **Practical:** standardise, initialise, monitor train/val loss, early stop, use schedules/momentum/Adam.



Nonconvex: convergence to stationary point, not guaranteed global optimum.

W13 NEURAL NETS

Neural network: composition of affine maps + nonlinear activations; learns representation.

Layer: $h^{(l)} = \phi_l(W^{(l)} h^{(l-1)} + b^{(l)})$; ReLU max(0, z). **Backprop:** compute loss backward, propagate gradients backward using chain rule, update by SGD/Adam. **Regularisation:** weight decay, dropout, early stopping, data augmentation, batch norm as stabiliser.

Choice	Effect	Risk
width/depth	capacity	overflow/compute
ReLU	sparse gradients	dead units
dropout	ensemble-like	slower train
softmax	class probs	calibration still needed

When to use: large data/unstructured/high interactions; for tabular small data, boosted trees often stronger baseline.

EXAM MICRO-RULES

Answer template: state method, formula/objective, assumptions, validation metric, failure mode, business implication.

Never claim test-set tuning, never use future labels; never optimise accuracy for rare costly events; always connect $\lambda, K, depth, \nu$ to bias-variance.

FORMULA BANK

Loss: squared $\frac{1}{2} \|y - \hat{y}\|^2$; abs $|y - \hat{y}|$; hinge max(0, $1 - y \hat{y}$); logloss $-\log p(y)$ + $(1-y) \log(1-p)$.

Risk: $R(f) = E[L(Y, f(X))]$, $\hat{R}(f) = n^{-1} \sum_i L_i$; validation estimates R , train estimates optimisation fit.

OLS derivation: minimise $\|y - X\beta\|^2$; normal eq $X^T X \hat{\beta} = X^T y$; projection $\hat{y} = X(X^T X)^{-1} X^T y$.

Ridge: minimise $\|y - X\beta\|^2 + \lambda \|\beta\|^2$; df shrinks as $\lambda \uparrow$; no exact zeros.

Lasso: minimise $\|y - X\beta\|^2 + \lambda \|\beta\|_1$; geometry corners $\Rightarrow$ sparsity; unstable among correlated twins.

Logistic: log $\frac{p}{1-p} = \eta$; $p = \sigma(\eta)$; decision boundary $\eta = 0$ for threshold .5; for threshold t , boundary $\eta = \log(t/(1-t))$.

Confusion: $TPR = TP/(TP + FN)$, $FPR = FP/(FP + TN)$, precision $= TP/(TP + FP)$, $F1 = 2PR/(P + R)$.

Bayes: $p(c|x) = p(c|x)p(c)/p(x)$; posterior odds = likelihood ratio $\times$ prior odds.

GINI: $G = 1 - \sum p_k^2$; entropy $H = -\sum p_k \log p_k$; regression split $SSE = \sum_{j \in \text{left}} \sum_{i \in R_j} (y_i - y_j)^2$.

Boosting: negative gradient target $r_{im} = -\partial L(y_i, F(x_i)) / \partial F(x_{im})$.

MLE: iid likelihood product $\prod_i p(y_i|x_i, \theta)$; log turns product to sum; Hessian curvature helps uncertainty if regular conditions hold.

Optim: GD $\theta^+ = \theta - \eta g$; Newton $\theta^+ = \theta - H^{-1} g$; SGD uses unbiased noisy g_B .

MODEL CHOOSER

Situation	Try	Why
small n , linear	ridge/lasso	stable/interp
sparse text	NB/logistic	high-df sparse
nonlinear 1D	GB/NN	smooth
tabular strong	RF/GBM	interactions
costly FN	calibrated prob-threshold	cost action
high p low n	regularised/feature select	variance control
time split	time split	feature realism
repeated event	group split	avoid entry leakage

ALGORITHM STEPS

CV tuning: choose grid/search $\rightarrow$ split train folds $\rightarrow$ fit preprocessing inside fold $\rightarrow$ train model $\rightarrow$ score held fold $\rightarrow$ average $\rightarrow$ refit best on train $\rightarrow$ final test once.

Valid path: standardise X $\rightarrow$ choose λ grid $\rightarrow$ CV $\rightarrow$ inspect selected variables/stability $\rightarrow$ refit/validate carefully $\rightarrow$ report predictive not causal interpretation.

Tree fit: start root $\rightarrow$ evaluate candidate splits $\rightarrow$ choose max impurity decrease $\rightarrow$ recurse until stopping $\rightarrow$ prune/tune $\rightarrow$ predict leaf average/class prob.

RF fit: for b trees, bootstrap rows, grow tree with $n \cdot \text{try}$ random features per split; average votes/probs; tune $m \cdot \text{try}$, min leaf, trees.

GBM fit: init constant F_0 ; compute pseudo-residuals; fit shallow tree; line search leaf values; shrink by ν ; repeat; early stop by validation.

MLE answer: specify distribution $\rightarrow$ write density/pmf $\rightarrow$ independence product $\rightarrow$ log-likelihood $\rightarrow$ differentiate or state numerical optimiser $\rightarrow$ interpret estimate.

SGD answer: objective minibatch estimate $\rightarrow$ gradient update $\rightarrow$ learning-rate schedule/momentum $\rightarrow$ validation monitoring $\rightarrow$ early stopping.

DIAGNOSTIC PATTERNS

Observation	Likely cause	Fix
train good val bad	variance/leakage	regularise, audit split
both bad	bias/features	richer reg/model
val swings by fold	unstable/small n	simpler model, repeated CV
good AUC bad profit	threshold/class	decision analysis
good val bad group	slice/class group	segment metrics
probs too extreme	miscalibration	Platt/isotonic
new data worse	drift	monitor/retrain
coeff signs weird	collinearity/leakage	ridge/audit

MINI EXAMPLES

Rare disease: prevalence .02, sens .90, spec .95. $P(D|+) = .9 \cdot (.02) / [.9 \cdot (.02) + .05 \cdot (.98)] \approx .269$; high TPR can still yield modest PPV.

Ridge vs lasso: if 80 predictors and 15 true signals, lasso may select; if predictors are correlated groups and prediction is goal, ridge/elastic net often safer.

Learning curve: Model D train .05 val .90 = overfit; Model B train .35 val .30 = more stable; do not prefer D by training error.

Threshold: if $C_{FN} = 4 C_{FP}$, predict positive when $p > C_{FP}/(C_{FP} + C_{FN}) = .2$ under calibrated probabilities.

COMMON TRAPS

Target leakage: post-disposition complaints, future churn reason, full-period aggregate, normalising using whole data, duplicate entry across splits.

Metric mismatch: accuracy for 1% positives, RMSE when outliers dominate and MAE is action-relevant, AUC when actual threshold/cost matters.

Interpretation: coefficient $\neq$ causal effect unless design supports it; tree importance $\neq$ stable causal importance; GAM partial curve conditional on

Regularisation: compare after scaling; λ chosen by CV; lasso selection not proof of true variable; ridge keeps all variables.

Uncertainty: CV mean without fold variability can mislead; model selected by min validation loss has selection bias; test set is final audit only.

Deployment: training distribution, labels, incentives, and action policy can shift;

monitor inputs, outputs, calibration, slices, and delayed outcomes.

DERIVATION SKELETONS

OLS from MLE: assume $y_i|x_i \sim N(x_i^T \beta, \sigma^2)$ iid. Then $\ell = C - \frac{1}{2\sigma^2} \sum_i (y_i - x_i^T \beta)^2$; maximise ℓ if minimise SSE; differentiate gives $X^T (y - X\beta) = 0$.

Logistic score: $\ell = \sum_i [y_i x_i^T \beta - \log(1 + \exp(x_i^T \beta))]$; $\nabla \ell = X^T (y - p)$; Hessian $-X^T W X$ so concave for unpenalised logistic.

Ridge MAP: prior $\beta \sim N(0, \tau^2 I)$ gives log posterior $-\frac{1}{2\sigma^2} \|y - X\beta\|^2 - \frac{\tau^2}{2} \|\beta\|^2$; $\lambda = \sigma^2/\tau^2$.

Bayes threshold: act positive if $C_{FP} P(Y=0|x) < C_{FN} P(Y=1|x)$; rearrange $p > C_{FP}/(C_{FP} + C_{FN})$.

Bias-var: for squared loss $E(p(x) - y)^2 = \sigma^2 + \text{Bias}(\hat{f})^2 + \text{Var}(\hat{f})$ at x ; biasing targets variance; regularisation can add bias to reduce variance.

EM/GMM if recalled: E-step responsibilities $r_{ik} \propto \pi_i N(x_i|\mu_k, \Sigma_k)$; M-step maximize means/covariances; increases lower bound, not necessarily global optimum.

PAST-PAPER CUES

If question says "compare algorithms": discuss assumptions, flexibility, interpretability, computation, tuning, data size, metric, failure mode.

If question gives imbalance data: mention baseline choice, confusion if model, PR-AUC/F1/recall, calibration, threshold by costs, stratified CV.

If question asks "best/worst classifier for plot": linear boundary with local boundary vs cluster Gaussian; KNN $K=1$ jagged; logistic rigid; QDA curved; ridge axis-aligned; boosting adaptive.

If question asks "feature selection": distinguish prediction vs explanation; best subset expensive/unstable; lasso sparse; ridge stable; use nested CV; post-selection inference risky.

If question asks "learning curve": high train error = bias; high val gap = variance; more data helps variance more than bias; more complexity helps bias but may increase variance.

If question asks likelihood: include constants unless told proportional; specify iid; identify posterior/likelihood/prior/inference for Bayes.

If question asks "ensemble": bagging parallel variance reduction; RF decorrelation via feature subsampling; boosting sequential residual/gradient correction.

METHOD PROS/CONS

Method	Pros	Cons
LDA	calibratable, interpretable	linear log-odds
Logistic	efficient if Gaussian	shared covariance
QDA	curved boundary	more params
simple/linear	simple	high dim
Tree	interpretable/interactions	unstable
QNN	strong default	less interpretable
GMM	high accuracy/tunable	overfitting labels
GBM	smooth explainable	weak interactions
NN	representation learning	data/compute/tuning

VALIDATION DESIGNS

Iid random split: okay if rows exchangeable and no entropy/time leakage.

Stratified: preserve class proportions. **Group:** all rows for a customer/product/patient in same fold. **Time:** train past, validate future.

Rolling-origin for repeated measures. **Time-series** for time series.

Nested: outer folds estimate performance; inner folds tune hyperparameters; prevents selection bias. **Repeated CV:** reduces noise when n small.

Bootstrap: can assess uncertainty but must respect groups/time. **Preprocess within resampling:** imputation, scaling, encoding, feature selection, target encoding. **Model** must be fit only on analysis fold.

BUSINESS TRANSLATION

Credit scoring: probability of default; costs asymmetric; calibration + threshold; reject inference/feedback risk.

Churn: prediction window vs intervention window; uplift may differ from churn risk; avoid features after cancellation.

Fraud: extreme imbalance, delayed labels, adaptive adversaries; high recall may create review burden.

Recommendation: ranking metrics, popularity bias, cold-start; hybrid collaborative + content features.